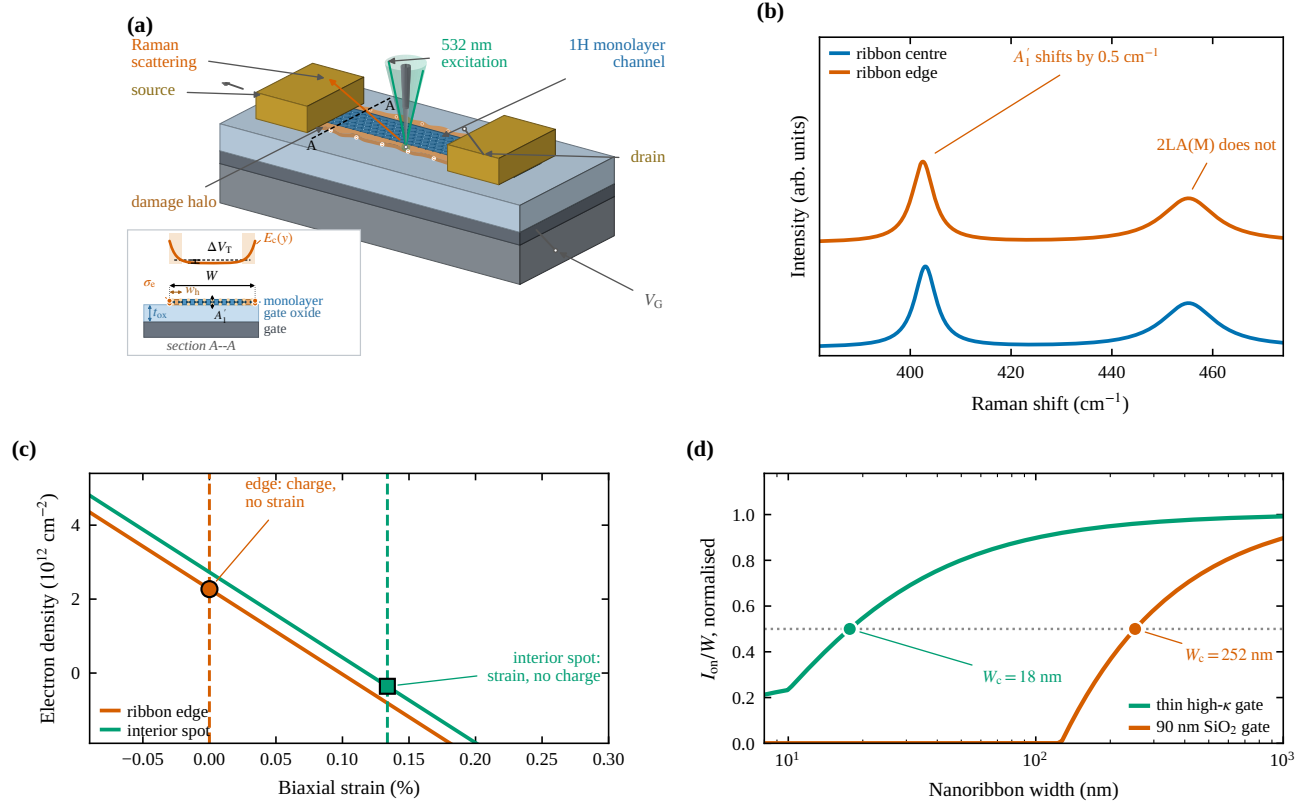

(a)
Raman scattering
source
532 nm excitation
1H monolayer channel
drain
damage halo
A
A
V_G
ΔV_T
$E_c(y)$
W
σ_e
w_h
monolayer
t_{ox}
A_1'
gate oxide
gate
section A--A
(b)
ribbon centre
ribbon edge
A_1' shifts by 0.5 cm^{-1}
2LA(M) does not
Intensity (arb. units)
400
420
440
460
Raman shift (cm^{-1})
(c)
edge: charge, no strain
interior spot: strain, no charge
ribbon edge
interior spot
Electron density (10^{12} cm^{-2})
4
2
0
−0.05
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Biaxial strain (%)
(d)
I_{on}/W, normalised
1.0
0.8
0.6
0.4
0.2
0.0
$W_c = 18$ nm
$W_c = 252$ nm
thin high-κ gate
90 nm SiO_2 gate
10^1
10^2
10^3
Nanoribbon width (nm)